Learning and computation in quantum systems

Premio Federico Tonielli 2026

Antonio Anna Mele

Freie Universität Berlin
Google PhD Fellow in Quantum Computing

Pisa, 16 ottobre 2026

My research path

2016–19 — **Pisa** — BSc in Physics

2019–21 — **Trento · SISSA** — MSc in Physics

2021– — **Berlin** — PhD

2024 — **Los Alamos** — Research internship

2025 Google Fellowship

2026 — **Google Quantum AI** — Research internship

Quantum information and computing

Quantum learning, quantum algorithms and noise, mathematical tools

Research map

Quantum learning

Qubits

[1] Optimal learning of quantum channels in diamond distance

Antonio Anna Mele, Lennart Bittel.

Nature Physics, accepted (2026)

[2] Quantum memory advantage for quantum process tomography

Carlos Bravo-Prieto, Weiyuan Gong, **Antonio Anna Mele**.

2026

[3]† Adaptivity is all you need: Optimal stabilizer learning using just single-copy measurements

Lennart Bittel, Jens Eisert, Weiyuan Gong, **Antonio Anna Mele**, Louis Schatzki.

2026

Fermions

[6] Efficient learning of quantum states prepared with few fermionic non-Gaussian gates

Antonio Anna Mele, Yaroslav Herasymenko.

PRX Quantum (2025)

[7] Optimal trace-distance bounds for free-fermionic states: Testing and improved tomography

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[8] Learning fermionic correlations by evolving with random translationally invariant Hamiltonians

Janek Denzler, **Antonio Anna Mele**, Ellen Derbyshire, Tommaso Guaita, Jens Eisert.

Physical Review Letters (2024)

[9] PAC-learning of free-fermionic states is NP-hard

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Quantum (2025)

[10] Testing quantum Gaussianity with constant sample complexity

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[4] Real classical shadows

Maxwell West, **Antonio Anna Mele**, Martin Larocca, Marco Cerezo.

Journal of Physics A (2025)

[5] Experimental measurement and a physical interpretation of quantum shadow enumerators

Daniel Miller, Kyano Levi, Lukas Postler, Alex Steiner, Lennart Bittel, Gregory A. L. White, Yifan Tang, Eric J. Kuehnke, **Antonio Anna Mele**, Sumeet Khatri, Lorenzo Leone, Jose Carrasco, Christian D. Marciniak, Ivan Pogorelov, Milena Guevara-Bertsch, Robert Freund, Rainer Blatt, Philipp Schindler, Thomas Monz, Martin Ringbauer, Jens Eisert.

Physical Review Research (2026)

Bosons

[11] Learning quantum states of continuous-variable systems

Francesco Anna Mele, **Antonio Anna Mele**, Lennart Bittel, Jens Eisert, Vittorio Giovannetti, Ludovico Lami, Lorenzo Leone, Salvatore F. E. Oliviero.

Nature Physics (2025)

[12] Optimal estimates of trace distance between bosonic Gaussian states and applications to learning

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Quantum algorithms and noise

Noisy circuits

[16] Noise-induced shallow circuits and the absence of barren plateaus

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Variational algorithms

[18] Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian variational ansatz

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Physical Review A (2022)

[19] Stochastic noise can be helpful for variational quantum algorithms

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[20] Exploiting symmetry in variational quantum machine learning

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PRX Quantum (2023)

Verifiable quantum advantage

[21] Towards verifiable quantum advantage with random circuits: Observables that survive concentration

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Mathematical tools and structure

Haar, Clifford and representation theory

[22] Introduction to Haar measure tools in quantum information: A beginner's tutorial

Antonio Anna Mele.

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[23]† A complete theory of the Clifford commutant

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Quantum (2026)

[24] Lecture Notes on Representation Theory for Quantum Information (QMATH Masterclass 2025, Copenhagen)

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Lecture notes (2026)

Randomness

[25] Strong unitary designs in optimal depth and space

Teodor Parella-Dilmé, Júlia Barberà-Rodríguez, Salvatore F. E. Oliviero, **Antonio Anna Mele**.

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Quantum resource theories

[27] A unified approach to quantum resource theories and a new class of free operations

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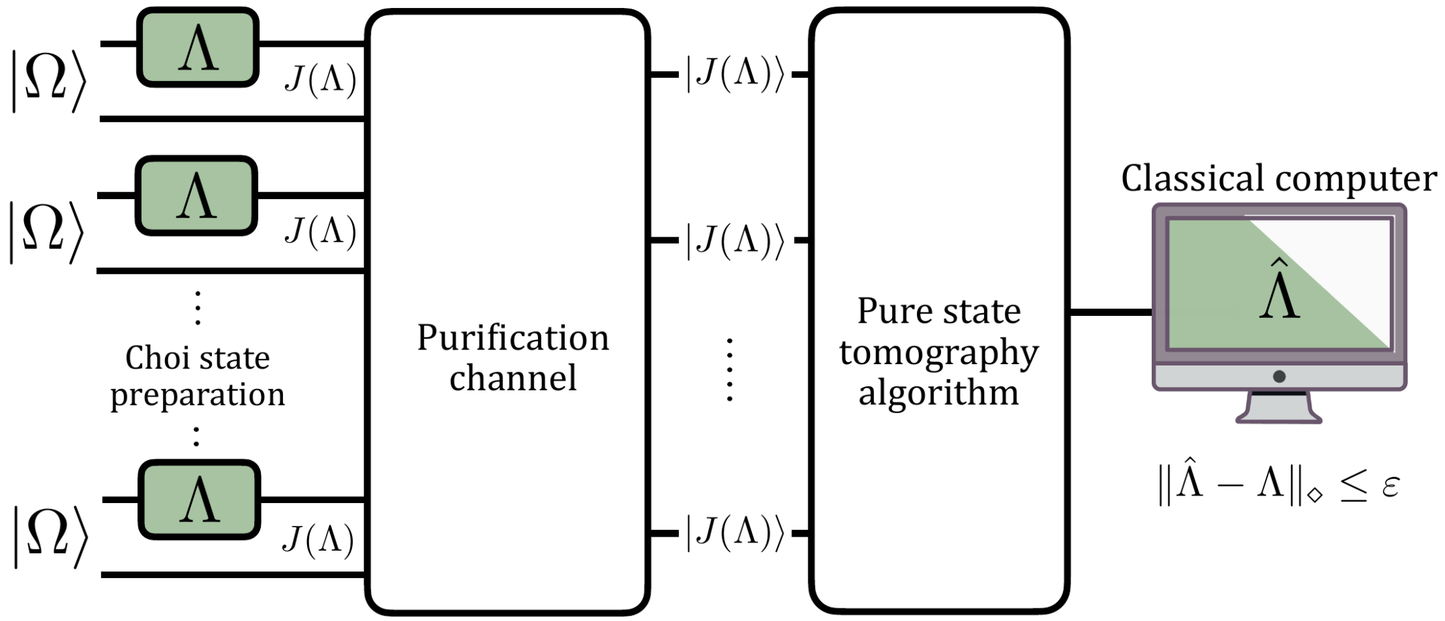
Advanced Quantum Technologies (2026)

* Equal contribution † Alphabetical author order

Optimal quantum process tomography

How many channel uses are needed to characterize a channel?

Choi state



Query complexity [1]

$$N = O\left(\frac{d_{\text{in}} d_{\text{out}} k}{\varepsilon^2}\right)$$

Kraus rank k

Diamond-distance guarantee

$$\|\widehat{\Lambda} - \Lambda\|_\diamond \leq \varepsilon$$

Worst-case guarantee, including entangled inputs

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[2] **Quantum memory advantage for quantum process tomography**
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Learning fermionic states beyond Gaussianity

Gaussian states [7]

Correlation matrices give a stable description of the state

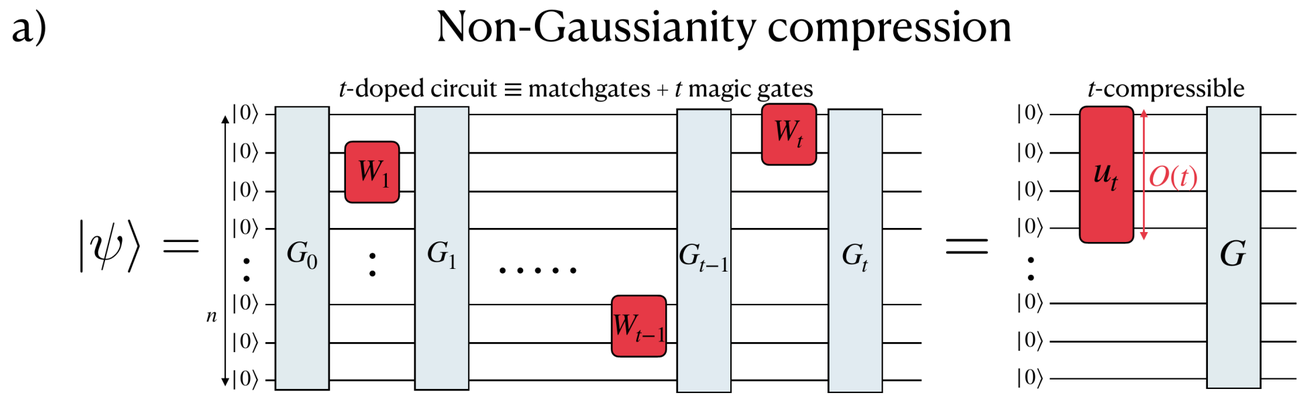
$$\Gamma_{jk} = -\frac{i}{2}\,\mathrm{Tr}\big(\rho[c_j, c_k]\big)$$

$$\|\rho - \sigma\|_1 \le \frac{1}{2}\|\Gamma(\rho) - \Gamma(\sigma)\|_1$$

Optimal trace-distance bounds for robust tomography

Few non-Gaussian gates [6]

Compress non-Gaussianity into a small subsystem



$$N,\ T_{\mathrm{cl}} = \mathrm{poly}(n, 2^t)$$

Single-copy measurements, fixed Majorana locality

Efficient when $t = O(\log n)$

[7] Optimal trace-distance bounds for free-fermionic states: Testing and improved tomography

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PRX Quantum 6, 030341 (2025).

[6] Efficient learning of quantum states prepared with few fermionic non-Gaussian gates

Antonio Anna Mele, Yaroslav Herasymenko.

PRX Quantum 6, 010319 (2025).

Learning bosonic states beyond Gaussianity

Gaussian structure enables efficient continuous-variable tomography [11]

Energy bounds alone do not ensure efficiency

$$N = \Omega(\varepsilon^{-2n})$$

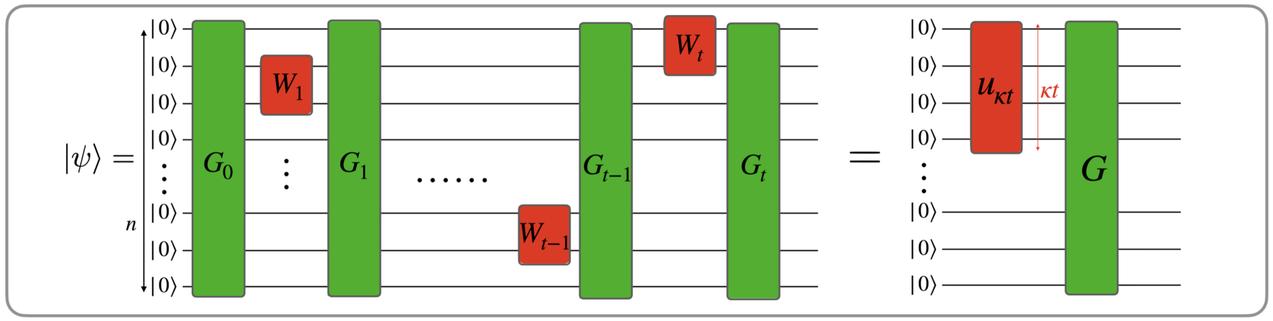
n modes with bounded mean energy per mode

Gaussian structure

First and second moments enable efficient trace-distance tomography

Beyond Gaussian states

Few local non-Gaussian gates
confine non-Gaussianity to a small subsystem



Efficient learning when $\kappa t = O(1)$

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Francesco Anna Mele, **Antonio Anna Mele**, Lennart Bittel, Jens Eisert, Vittorio Giovannetti, Ludovico Lami, Lorenzo Leone, Salvatore F. E. Oliviero.
Nature Physics 21, 2002–2008 (2025).

Other works in learning bosonic systems

Trace-distance bounds [12]

Optimal trace-distance estimates
from errors in first and second moments

Gaussian unitaries [15]†

Learning transformations with an
energy-constrained diamond guarantee

Energy-independent tomography [13]

Adaptive unsqueezing gives a double-exponential improvement in energy dependence over previous protocols.

Gaussianity testing [10]

(Distinguishing Gaussian from non-Gaussian)

For pure inputs: a number of copies independent of system size

[12] Optimal estimates of trace distance between bosonic Gaussian states and applications to learning

Lennart Bittel, Francesco Anna Mele, **Antonio Anna Mele**, Salvatore Tirone, Ludovico Lami.

Quantum 9, 1769 (2025).

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Separations in learning

Quantum channels [2]

Measure after every query, or retain quantum memory

$$N_{\text{classical memory}} = \Theta\left(D^3/\varepsilon^2\right)$$

$$N_{\text{quantum memory}} = \Theta\left(D^2/\varepsilon^2\right)$$

$$D = d_{\text{in}} d_{\text{out}}$$

Stabilizer states [3]†

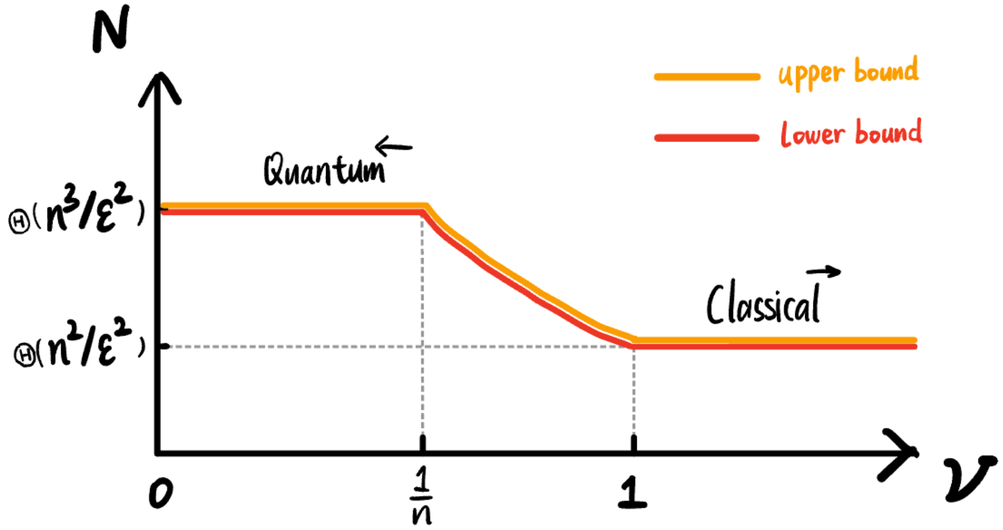
$$N_{\text{nonadaptive, single-copy}} = \Omega(n^2)$$

$$N_{\text{adaptive, single-copy}} = \Theta(n)$$

Bosonic Gaussian states [14]†

$$N^{\text{cold}}_{\text{single-copy}} = \Omega(n^3), \quad N^{\text{cold}}_{\text{collective}} = \Theta(n^2)$$

$$N^{\text{warm}}_{\text{heterodyne}} = N^{\text{warm}}_{\text{collective}} = \Theta(n^2)$$



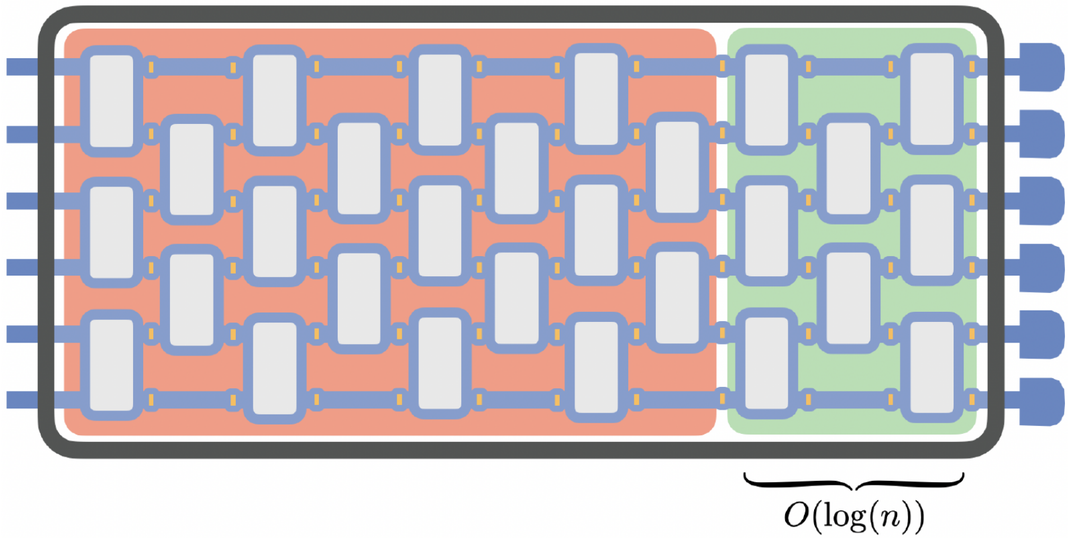
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Local noise induces an effective shallow circuit

For typical random circuits, local noise limits the relevant depth



Exponential forgetting [16]

$$\mathbb{E}_C |\langle O \rangle_{\text{full}} - \langle O \rangle_{\text{last } m}| \leq \|O\|_\infty e^{-\alpha m}$$

$$L_{\text{eff}} = O(\log n)$$

Fixed local noise,
inverse-polynomial accuracy

Classical simulation [17]

Pauli propagation with controlled truncation

$$T = \text{poly}(n, \varepsilon^{-1}, \delta^{-1})$$

[16] Noise-induced shallow circuits and the absence of barren plateaus

Antonio Anna Mele, Armando Angrisani, Soumik Ghosh, Sumeet Khatri, Jens Eisert, Daniel Stilck França, Yihui Quek.

Nature Physics 22, 751–756 (2026).

[17] Simulating quantum circuits with arbitrary local noise using Pauli propagation

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PRX Quantum 7, 020313 (2026).

Out-of-time-order correlators (OTOCs)

An OTOC quantifies how a local perturbation B affects a distant operator M.

Task: estimate the OTOC

Random circuit U, prepared state ρ,
distant local Pauli operators B and M

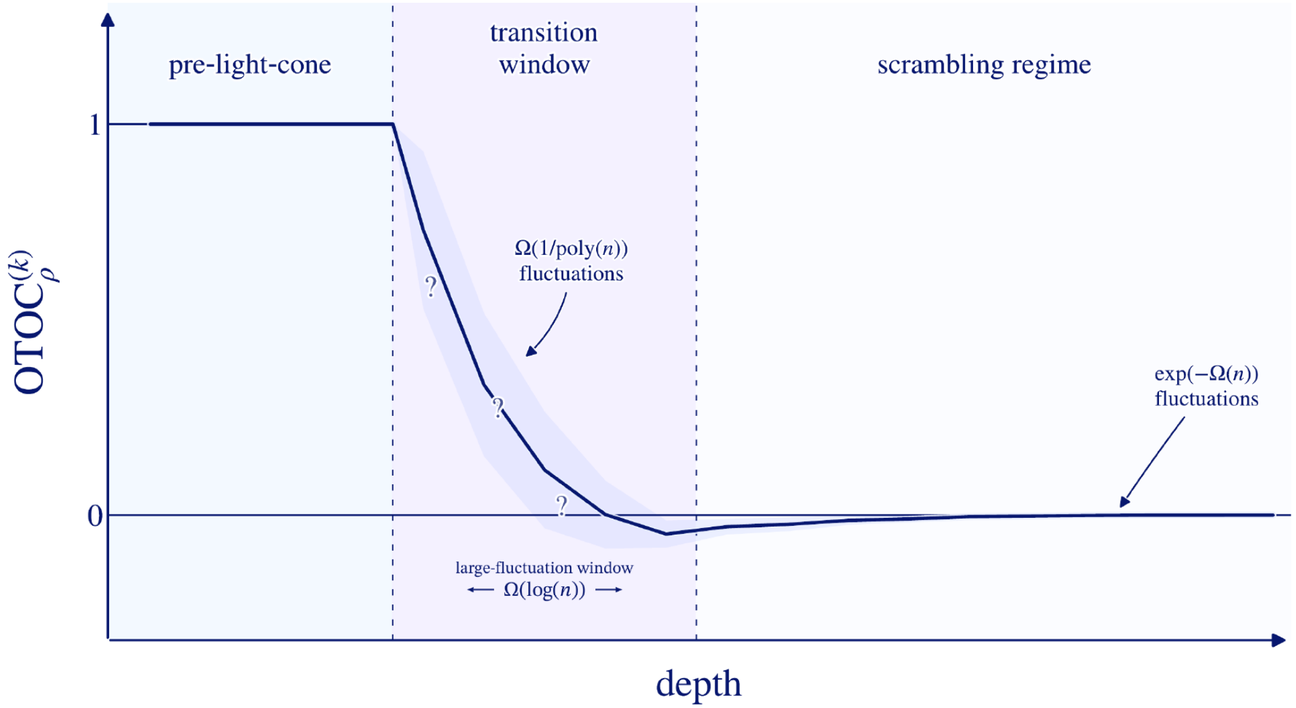
$$B_U = U^\dagger B U$$

$$\mathrm{OTOC}_\rho^{(k)}(U) = \mathrm{Tr}\left[\rho (B_U M)^{2k}\right]$$

$$\left|\widehat{z} - \mathrm{OTOC}_\rho^{(k)}(U)\right| \le \varepsilon, \quad \varepsilon = \frac{1}{\mathrm{poly}(n)}$$

Fluctuations survive at intermediate depth

$$\mathrm{Var}_U\left[\mathrm{OTOC}_\rho^{(k)}(U)\right] \ge \frac{1}{\mathrm{poly}(n)}$$



Collaboration with Google Quantum AI

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* Equal contribution

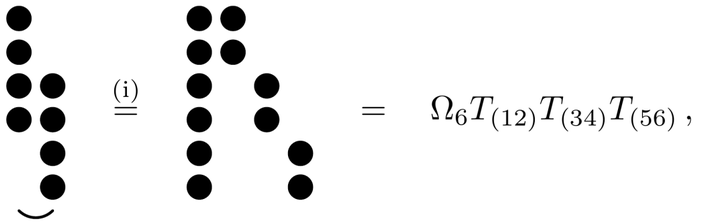
Mathematical tools and structure

Clifford commutant [23]†

Higher-order symmetries of the qubit Clifford group

$$[O, C^{\otimes k}] = 0 \quad \forall C \in \mathcal{C}_n$$

QEC and magic-state resource theory



$$= \Omega_6 T_{(12)} T_{(34)} T_{(56)} \,,$$

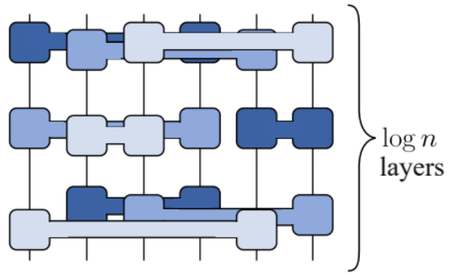
Haar and representation theory

Haar integration, moment operators and representation theory lecture notes [22, 24]

Strong unitary designs [25]

$$d = \Theta(\log n) \quad (n \text{ qubits})$$

All-to-all connectivity



[22] Introduction to Haar measure tools in quantum information: A beginner's tutorial
Antonio Anna Mele. ***Quantum*** 8, 1340 (2024).

[23]† A complete theory of the Clifford commutant
Lennart Bittel, Jens Eisert, Lorenzo Leone, **Antonio Anna Mele**, Salvatore F. E. Oliviero. ***Quantum*** 10, 2171 (2026).

[24] Lecture Notes on Representation Theory for Quantum Information (QMATH Masterclass 2025, Copenhagen)
Antonio Anna Mele. Lecture notes (2026).

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Teaching and mentoring

QMATH, Copenhagen · August 2025

Representation theory crash course

AIMS, Cape Town · October 2025

Haar integration tools

Freie Universität Berlin · 2022–23

Quantum information theory

Teaching assistant

QISCA Winter School · January 2026

Learning t-doped Gaussian states

EPFL Bernoulli Center · May 2026

Bosonic and fermionic quantum learning

MSc mentoring · Diego Polimeni · 2025

Fault-tolerant quantum computation

Quantum algorithms

Fault-tolerant quantum computing

AI-powered research

Thank you so much for your attention

Antonio Anna Mele

antonioannamele.com

Fermions

[6] **Antonio Anna Mele**, Yaroslav Herasymenko.
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PRX Quantum 6, 010319 (2025)

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Learning fermionic correlations by evolving with random translationally invariant Hamiltonians
Physical Review Letters 133, 240604 (2024)

[9] Lennart Bittel, **Antonio Anna Mele**, Jens Eisert, Lorenzo Leone.
PAC-learning of free-fermionic states is NP-hard
Quantum 9, 1665 (2025)

[10] Mahtab Yaghubi Rad, Ricard Puig, Saksham Hassanandani, Luke Coffman, Jens Eisert, Carlos Bravo-Prieto, **Antonio Anna Mele**.
Testing quantum Gaussianity with constant sample complexity
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Journal of Physics A: Mathematical and Theoretical 58, 245304 (2025)

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Experimental measurement and a physical interpretation of quantum shadow enumerators
Physical Review Research 8, 023318 (2026)

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Lecture notes (2026)

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